

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:10:18

Source: data/json/noise_experiments/v2/noise2i_match8_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_match8/noise2i_match8_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_match8
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$7; payoffs Agent_1=\$8, Agent_2=\$5;

noise/visible: sent visible=\$4.28, received visible=\$12.84, returned visible=\$7.94

Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 8.0,

"Agent_2": 5.0

}

Visible balances after round: {

"Agent_1": 8.940000000000001,

"Agent_2": 5.84

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```
'''json
{"send": 4}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 624,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:06:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ]
  }
}
```

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Assistant output:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: \$0.72 + \$7.00 = \$7.72
- My total: \$12.84 - \$7.00 = \$5.84

This seems fair and reciprocates their generous trust.

```
{"return": 7.00}
```

Reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $\$4.28 \times 3 = \12.84
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $\$5 - \$4.28 = \$0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $\$0.72 + \$6.42 = \$7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$1.5; payoffs Agent_2=\$2.5, Agent_1=\$10.5;
noise/visible: sent visible=\$0.77, received visible=\$2.31, returned visible=\$1.03
Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 12.0, "min": 0, "raw": 1.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 18.5,
"Agent_2": 7.5

}

Visible balances after round: {

"Agent_1": 9.750000000000002,
"Agent_2": 7.87

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: $\$0.72 + \$7.00 = \$7.72$
- My total: $\$12.84 - \$7.00 = \$5.84$

This seems fair and reciprocates their generous trust.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $\$4.28 \times 3 = \12.84
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $\$5 - \$4.28 = \$0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $\$0.72 + \$6.42 = \$7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust

- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

Reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: \$1.00 + \$6.50 = \$7.50

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: \$0.50 + \$7.00-\$7.50 = \$7.50-\$8.00

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

Usage: {

```
"input_tokens": 609,  
"output_tokens": 638,  
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
  "investor": "Agent_2",
```

```

"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```

'''json
{"send": 4}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 624,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Assistant output:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json
{"return": 1.50}
'''
```

Reasoning:

Let me think about this situation. I'm now the RECEIVER.

In Round 1, I was the SENDER and sent \$4. That would have been tripled to \$12 for the receiver

Now in Round 2, I'm the receiver and the sender sent me \$0.77, which was tripled to \$2.31.

My total earnings across all rounds are \$8.94. Let me work backwards:

- Round 1: I sent \$4, so I had \$5 - \$4 = \$1 left, plus whatever was returned to me.
- If my total is \$8.94, and I started with \$1 from Round 1, that means I must have received \$7.94 back from the receiver in Round 1.

Wait, let me recalculate:

- Round 1: I sent \$4, which became \$12 for the receiver
- The receiver returned some amount to me
- I kept: \$5 - \$4 + returned = \$1 + returned
- My current total is \$8.94
- So: \$1 + returned = \$8.94
- Therefore: returned = \$7.94

That's interesting - the receiver returned \$7.94 out of \$12, which is about 66%. That's quite generous!

Now in Round 2, the sender sent me \$0.77 (which seems like an odd amount - might be due to the communication noise mentioned). This tripled to \$2.31.

Given that the previous receiver was quite generous (returning about 66%), I should reciprocate that level of cooperation. This sender sent me \$0.77, which is a relatively small amount compared to my \$4 in Round 1.

If I apply a similar return rate of ~66%, I would return about $0.66 * \$2.31 = \1.52 .

The sender's smaller contribution suggests less confidence, but still demonstrates willingness to participate....

```
Usage: {
  "input_tokens": 537,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$0.5; payoffs Agent_1=\$1.5, Agent_2=\$11.5;

noise/visible: sent visible=\$0.5, received visible=\$1.5, returned visible=\$0.32

Action validation: {"returned": {"amount": 0.5, "clamped": false, "max": 12.0, "min": 0, "raw": 0.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 20.0,
"Agent_2": 19.0

}

Visible balances after round: {

"Agent_1": 11.070000000000002,
"Agent_2": 8.870000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```

'''json
{"send": 4}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optim

al strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 624,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json
{"return": 1.50}
'''
```

reasoning:

Let me think about this situation. I'm now the RECEIVER.

- That's about 66% return rate - very generous!

Round 2: I was RECEIVER

- Sender sent \$0.77
- I received \$2.31 (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

```
Usage: {  
  "input_tokens": 699,  
  "output_tokens": 503,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: \$0.72 + \$7.00 = \$7.72
- My total: \$12.84 - \$7.00 = \$5.84

This seems fair and reciprocates their generous trust.

{"return": 7.00}

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $4.28 \times 3 = 12.84$
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $5 - 4.28 = 0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $0.72 + 6.42 = 7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):
- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:
- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{  
  "input_tokens": 394,  
  "output_tokens": 875,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: $\$1.00 + \$6.50 = \$7.50$

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: $\$0.50 + \$7.00 - \$7.50 = \$7.50 - \$8.00$

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87.

This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Assistant output:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

```
{"return": 0.50}
```

Reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

Given the low trust shown, I need to consider:

- If I return too much, I lose significantly from the small amount I received
- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

A proportional return might be appropriate. The sender sent 10% of their endowment. If I return a similar proportion of what I received, that would be about \$0.15 (10% of \$1.50).

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

If I return \$0.50:
- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain \$1.00 this round)

If I return \$0.75:
- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain \$0.75 this round)

```
...
Usage: {
  "input_tokens": 754,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_1 returned \$0.15; payoffs Agent_2=\$2.15,
Agent_1=\$8.85; noise/visible: sent visible=\$0.09, received visible=\$0.27, returned visible=\$0.65
Action validation: {"returned": {"amount": 0.15, "clamped": false, "max": 9.0, "min": 0, "raw": 0.15}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 28.85,
"Agent_2": 21.15

}

Visible balances after round: {

"Agent_1": 11.190000000000001,
"Agent_2": 11.520000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"

}

"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: \$0.72 + \$7.00 = \$7.72
- My total: \$12.84 - \$7.00 = \$5.84

This seems fair and reciprocates their generous trust.

{"return": 7.00}

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $4.28 \times 3 = 12.84$
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $5 - 4.28 = 0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $0.72 + 6.42 = 7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding

the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: $\$1.00 + \$6.50 = \$7.50$

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: $\$0.50 + \$7.00 - \$7.50 = \$7.50 - \$8.00$

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87. This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain in some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

```
{"return": 0.50}
```

reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

Given the low trust shown, I need to consider:

- If I return too much, I lose significantly from the small amount I received

- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

A proportional return might be appropriate. The sender sent 10% of their endowment. If I return a similar proportion of what I received, that would be about \$0.15 (10% of \$1.50).

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

If I return \$0.50:

- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain $\$1.00$ this round)

If I return \$0.75:

- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain $\$0.75$ this round)

• • •

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

You are the SENDER this round. Your total earnings across all rounds are \$8.87000000000000

1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience across the rounds:

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

Given that I was somewhat exploited in Round 2 when I sent a large amount, I should be more cautious this round. However, I'll still send a moderate amount to test cooperation.

```
{"send": 3.00}
```

Reasoning:

Let me analyze the pattern so far:

Round 1 (I was RECEIVER):

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

Round 2 (I was SENDER):

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- $\$7.87 = \$5.84 + \$1.00 + \text{returned}$
- $\$7.87 = \$6.84 + \text{returned}$
- $\text{returned} = \$1.03$

So the receiver only returned \$1.03 out of \$12.00! That's only about 8.6%! That's very low reciprocity.

Round 3 (I was RECEIVER):

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50

- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- $\$7.87 + \$1.00 = \$8.87$?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

```
Usage: {
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
```

```
"input_tokens": 365,  
"output_tokens": 624,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json  
{"return": 1.50}  
'''
```

reasoning:

Let me think about this situation. I'm now the RECEIVER.

In Round 1, I was the SENDER and sent \$4. That would have been tripled to \$12 for the receiver.

Now in Round 2, I'm the receiver and the sender sent me \$0.77, which was tripled to \$2.31.

My total earnings across all rounds are \$8.94. Let me work backwards:

- Round 1: I sent \$4, so I had \$5 - \$4 = \$1 left, plus whatever was returned to me.
- If my total is \$8.94, and I started with \$1 from Round 1, that means I must have received \$7.94 back from the receiver in Round 1.

Wait, let me recalculate:

- Round 1: I sent \$4, which became \$12 for the receiver
- The receiver returned some amount to me
- I kept: \$5 - \$4 + returned = \$1 + returned
- My current total is \$8.94
- So: \$1 + returned = \$8.94
- Therefore: returned = \$7.94

That's interesting - the receiver returned \$7.94 out of \$12, which is about 66%. That's quite generous!

Now in Round 2, the sender sent me \$0.77 (which seems like an odd amount - might be due to the communication noise mentioned). This tripled to \$2.31.

Given that the previous receiver was quite generous (returning about 66%), I should reciprocate that level of cooperation. This sender sent me \$0.77, which is a relatively small amount compared to my \$4 in Round 1.

If I apply a similar return rate of ~66%, I would return about $0.66 * \$2.31 = \1.52 .

The sender's smaller contribution suggests less confidence, but still demonstrates willingness to participate....

usage:

```
{  
  "input_tokens": 537,  
  "output_tokens": 703,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.7500000000000000002.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$4 and received a very generous return (~\$7.94)

Round 2: The sender sent \$0.77, and I returned \$1.50 (about 65%)

The receiver has shown strong cooperation. Even though they sent less in Round 2 (possibly affected by the noise mentioned in the instructions), we've established a pattern of mutual trust.

I'll continue to cooperate by sending a substantial amount again.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track what's happened:

Round 1: I was SENDER

- I sent \$4
- The receiver got \$12 (4×3)
- My total earnings are \$8.94
- So I kept $\$5 - \$4 = \$1$, and received back some amount
- $\$1 + \text{returned} = \8.94
- So I received \$7.94 back from the \$12 they got
- That's about 66% return rate - very generous!

Round 2: I was RECEIVER

- Sender sent \$0.77
- I received \$2.31 (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.0700000000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to the sender? (0-0.27)

Assistant output:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

Reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop. Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

```
Usage: {
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0; payoffs Agent_1=\$2, Agent_2=\$9;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.7

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 30.85,

"Agent_2": 30.15

}

Visible balances after round: {

"Agent_1": 13.89,

"Agent_2": 11.520000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:08:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 624,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json
{"return": 1.50}
'''
```

reasoning:

Let me think about this situation. I'm now the RECEIVER.

In Round 1, I was the SENDER and sent \$4. That would have been tripled to \$12 for the receiver.

Now in Round 2, I'm the receiver and the sender sent me \$0.77, which was tripled to \$2.31.

My total earnings across all rounds are \$8.94. Let me work backwards:

- Round 1: I sent \$4, so I had \$5 - \$4 = \$1 left, plus whatever was returned to me.
- If my total is \$8.94, and I started with \$1 from Round 1, that means I must have received \$7.94 back from the receiver in Round 1.

Wait, let me recalculate:

- Round 1: I sent \$4, which became \$12 for the receiver
- The receiver returned some amount to me
- I kept: \$5 - \$4 + returned = \$1 + returned
- My current total is \$8.94
- So: \$1 + returned = \$8.94
- Therefore: returned = \$7.94

That's interesting - the receiver returned \$7.94 out of \$12, which is about 66%. That's quite generous!

Now in Round 2, the sender sent me \$0.77 (which seems like an odd amount - might be due to the communication noise mentioned). This tripled to \$2.31.

Given that the previous receiver was quite generous (returning about 66%), I should reciprocate that level of cooperation. This sender sent me \$0.77, which is a relatively small amount compared to my \$4 in Round 1.

If I apply a similar return rate of ~66%, I would return about $0.66 * \$2.31 = \1.52 .

The sender's smaller contribution suggests less confidence, but still demonstrates willingness to participate....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.7500000000000002.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$4 and received a very generous return (~\$7.94)

Round 2: The sender sent \$0.77, and I returned \$1.50 (about 65%)

The receiver has shown strong cooperation. Even though they sent less in Round 2 (possibly affected by the noise mentioned in the instructions), we've established a pattern of mu

tual trust.

I'll continue to cooperate by sending a substantial amount again.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track what's happened:

Round 1: I was SENDER

- I sent \$4
- The receiver got \$12 (4×3)
- My total earnings are \$8.94
- So I kept $\$5 - \$4 = \$1$, and received back some amount
- $\$1 + \text{returned} = \8.94
- So I received \$7.94 back from the \$12 they got
- That's about 66% return rate - very generous!

Round 2: I was RECEIVER

- Sender sent \$0.77
- I received \$2.31 (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to The sender? ($0 - 0.27$)

Message 9 role=assistant

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop.

Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.190000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)

Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50

Round 3: I sent \$4

Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I'll send \$3 instead of \$4 this round.

```
Usage: {  
  "input_tokens": 1070,  
  "output_tokens": 604,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: \$0.72 + \$7.00 = \$7.72
- My total: \$12.84 - \$7.00 = \$5.84

This seems fair and reciprocates their generous trust.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $4.28 \times 3 = \$12.84$
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $5 - 4.28 = \$0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $0.72 + 6.42 = \$7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90%

would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: $\$1.00 + \$6.50 = \$7.50$

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: $\$0.50 + \$7.00 - \$7.50 = \$7.50 - \$8.00$

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87.

This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain in some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

{"return": 0.50}

reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

- If I return too much, I lose significantly from the small amount I received
- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain \$1.00 this round)

- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain $\$0.75$ this round)

```
{
  "input_tokens": 754,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

How much do you send to the receiver this round? (0-5)

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- \$7.87 = \$5.84 + \$1.00 + returned
- \$7.87 = \$6.84 + returned
- returned = \$1.03

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- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50
- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- $\$7.87 + \$1.00 = \$8.87$?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

usage:

```
{
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

Reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to return \$0 since I have \$0 to return from this round's transaction.

Usage: {

```
"input_tokens": 1178,
"output_tokens": 112,
"reasoning_tokens": 0
}
```

=====

ROUND 6

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $2; received $6; Agent_1 returned $0; payoffs Agent_2=$3, Agent_1=$6;
noise/visible: sent visible=$0, received visible=$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
```

Balances after round: {

```
"Agent_1": 36.85,
"Agent_2": 33.15
}
```

Visible balances after round: {

```
"Agent_1": 13.89,
"Agent_2": 14.520000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:08:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: $\$0.72 + \$7.00 = \$7.72$
- My total: $\$12.84 - \$7.00 = \$5.84$

This seems fair and reciprocates their generous trust.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $\$4.28 \times 3 = \12.84
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $\$5 - \$4.28 = \$0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $\$0.72 + \$6.42 = \$7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment

- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: \$1.00 + \$6.50 = \$7.50

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: \$0.50 + \$7.00-\$7.50 = \$7.50-\$8.00

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87.

This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50

Given that I was somewhat exploited in Round 2 when I sent a large amount, I should be more cautious this round. However, I'll still send a moderate amount to test cooperation.

```
{"send": 3.00}
```

reasoning:

Let me analyze the pattern so far:

Round 1 (I was RECEIVER):

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

Round 2 (I was SENDER):

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- \$7.87 = \$5.84 + \$1.00 + returned
- \$7.87 = \$6.84 + returned
- returned = \$1.03

So the receiver only returned \$1.03 out of \$12.00! That's only about 8.6%! That's very low reciprocity.

Round 3 (I was RECEIVER):

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50
- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- \$7.87 + \$1.00 = \$8.87 ?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

usage:

```
{
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to re

turn \$0 since I have \$0 to return from this round's transaction.

usage:

```
{
  "input_tokens": 1178,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.520000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my track record as sender:

- Round 2: I sent \$4.00 (80%) and received minimal return - heavily exploited
- Round 4: I sent \$3.00 (60%) and again received very little back - poor reciprocity

The last round someone sent me \$0, suggesting cooperation has broken down. However, this might be a different receiver, and I should give some opportunity for cooperation while protecting myself from further exploitation.

```
{"send": 2.00}
```

Reasoning:

Let me review my history:

Round 1 (RECEIVER): Sender sent \$4.28 (85.6%), I received \$12.84, I returned \$7.00

Round 2 (SENDER): I sent \$4.00, I received back approximately \$1.03 (based on my earnings going from \$5.84 to \$7.87, the increase was \$2.03, but I had \$1 left after sending \$4, so \$1 + return = \$2.03, meaning I got back about \$1.03 from \$12 sent to receiver - very poor reciprocity)

Round 3 (RECEIVER): Sender sent \$0.50, I received \$1.50, I returned \$0.50

Round 4 (SENDER): I sent \$3.00, my earnings went from \$8.87 to \$11.52, that's an increase of \$2.65. I had \$2 left after sending \$3, so \$2 + return = \$2.65, meaning I got \$0.65 back from the \$9 the receiver got - again very poor reciprocity

Round 5 (RECEIVER): Sender sent \$0, I received \$0, I returned \$0

Pattern analysis:

- When I've been the sender (rounds 2 and 4), I've shown trust but received very little back
- Round 2: I sent \$4 (80%), got back only about \$1.03 from \$12 (8.6% return rate)
- Round 4: I sent \$3 (60%), got back only about \$0.65 from \$9 (7.2% return rate)
- The receivers I've faced have been exploitative

Given this pattern of exploitation when I send money, and that in Round 5 someone just sent me \$0, cooperation seems to have broken down. I should be cautious and send less.

However, I also need to consider:

1. There's communication noise mentioned

...

Usage: {

```
"input_tokens": 1267,
"output_tokens": 651,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
```

```

],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```

'''json
{"send": 4}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optim

al strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 624,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json
{"return": 1.50}
'''
```

reasoning:

Let me think about this situation. I'm now the RECEIVER.

- So I received \$7.94 back from the \$12 they got
- That's about 66% return rate - very generous!

Round 2: I was RECEIVER

- Sender sent \$0.77
- I received \$2.31 (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to The sender? ($0 - 0.27$)

Message 9 role=assistant

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop.

Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.190000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)

Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50

Round 3: I sent \$4

Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I'll send \$3 instead of \$4 this round.

usage:

```
{
  "input_tokens": 1070,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I must return \$0.

This continuing downward trend (from \$0.77 to \$0.09 to \$0) is concerning, though it may be due to the communication noise mentioned in the instructions rather than the sender's actual choices.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return in this case.

This is interesting - the pattern shows:

- When I was sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I was receiver (Rounds 2, 4, 6): I received \$0.77, \$0.09, \$0

There's a clear downward trend in what the sender is sending me. This could be:

1. Communication noise (as mentioned in the instructions)
2. The sender is actually reducing their cooperation
3. Some combination

Since I received \$0, there's nothing I can return. The only valid response is to return \$0.

My total earnings of \$13.89 after 5 rounds is still positive, so the game has been somewhat profitable overall.

Usage: {

```
"input_tokens": 1269,
"output_tokens": 286,
"reasoning_tokens": 0
```

}

=====

ROUND 7

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$1; received \$3; Agent_2 returned \$0; payoffs Agent_1=\$4, Agent_2=\$3;

```

noise/visible: sent visible=$0, received visible=$0, returned visible=$0.21
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
Balances after round: {
  "Agent_1": 40.85,
  "Agent_2": 36.15
}
Visible balances after round: {
  "Agent_1": 18.1,
  "Agent_2": 14.520000000000001
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation and trust while being somewhat cautious since I don't have any history with this receiver yet.

If I send money, it triples in value, creating potential gains for both of us. However, I'm relying on the receiver to return a fair amount.

I'll send a substantial amount to show good faith and maximize our joint benefit, but keep a small buffer.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

There's also mentioned "communication noise" which means the amounts might be perceived differently by each player.

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd end up with \$7.50
- The socially optimal outcome is for me to send the maximum amount since it triples, maximizing total welfare
- But this requires trust that the receiver will return a fair amount

This is the first round, so I don't have any history with this receiver. Common strategies:

1. Send nothing (zero trust/risk averse)
2. Send a moderate amount to test trustworthiness
3. Send everything (maximize joint payoff, hoping for reciprocity)

Since this is a multi-round game and there's potential for learning and building trust, I think sending a moderate to high amount makes sense. This signals trust and cooperation, which might encourage the receiver to reciprocate.

I'll send \$3. This shows willingness to cooperate while not risking everything. The receiver will get \$9, and if they're fair-minded, they might return around \$4-5, leaving us both better off than if I sent nothing.

Actually, let me reconsider. The instructions mention this might involve multiple rounds with potentially the same or different partners. Building a reputation for cooperation could be beneficial. I'll send \$4, which shows strong trust while keeping \$1 as insurance.

Wait, let me think more carefully. The receiver gets \$12 if I send \$4. ...

usage:

```
{
  "input_tokens": 365,
```

```

        "output_tokens": 624,
        "reasoning_tokens": 0
    }
Message 4 role=user
    content:
        Round 2

        You are the RECEIVER this round. Your total earnings across all rounds are $8.940000000000000001.
        This round, The sender sent you $0.77, so you received $2.31. How much do you return to The sender? (0-2.31)
Message 5 role=assistant
    content:
        Looking at this situation:

        In Round 1, I sent $4 and received a generous return. Now the sender has sent me $0.77 (which became $2.31 after tripling).

        While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

        '''json
        {"return": 1.50}
        '''
    reasoning:
        Let me think about this situation. I'm now the RECEIVER.

        In Round 1, I was the SENDER and sent $4. That would have been tripled to $12 for the receiver.

        Now in Round 2, I'm the receiver and the sender sent me $0.77, which was tripled to $2.31.

        My total earnings across all rounds are $8.94. Let me work backwards:
        - Round 1: I sent $4, so I had $5 - $4 = $1 left, plus whatever was returned to me.
        - If my total is $8.94, and I started with $1 from Round 1, that means I must have received $7.94 back from the receiver in Round 1.

        Wait, let me recalculate:
        - Round 1: I sent $4, which became $12 for the receiver
        - The receiver returned some amount to me
        - I kept: $5 - $4 + returned = $1 + returned
        - My current total is $8.94
        - So: $1 + returned = $8.94
        - Therefore: returned = $7.94

        That's interesting - the receiver returned $7.94 out of $12, which is about 66%. That's quite generous!

        Now in Round 2, the sender sent me $0.77 (which seems like an odd amount - might be due to the communication noise mentioned). This tripled to $2.31.

        Given that the previous receiver was quite generous (returning about 66%), I should reciprocate that level of cooperation. This sender sent me $0.77, which is a relatively small amount compared to my $4 in Round 1.

        If I apply a similar return rate of ~66%, I would return about 0.66 * $2.31 = $1.52.
        The sender's smaller contribution suggests less confidence, but still demonstrates willingness to participate....
    usage:
        {
            "input_tokens": 537,
            "output_tokens": 703,
            "reasoning_tokens": 0
        }
Message 6 role=user
    content:
        Round 3

        You are the SENDER this round. Your total earnings across all rounds are $9.750000000000000000

```

2.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$4 and received a very generous return (~\$7.94)

Round 2: The sender sent \$0.77, and I returned \$1.50 (about 65%)

The receiver has shown strong cooperation. Even though they sent less in Round 2 (possibly affected by the noise mentioned in the instructions), we've established a pattern of mutual trust.

I'll continue to cooperate by sending a substantial amount again.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track what's happened:

Round 1: I was SENDER

- I sent \$4
- The receiver got \$12 (4×3)
- My total earnings are \$8.94
- So I kept $\$5 - \$4 = \$1$, and received back some amount
- $\$1 + \text{returned} = \8.94
- So I received \$7.94 back from the \$12 they got
- That's about 66% return rate - very generous!

Round 2: I was RECEIVER

- Sender sent \$0.77
- I received \$2.31 (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to The sender? (0-0.27)

Message 9 role=assistant

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop.

Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.1900000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty.

rtainty:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)

Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50

Round 3: I sent \$4

Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I'll send \$3 instead of \$4 this round.

usage:

```
{
  "input_tokens": 1070,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I must return \$0.

This continuing downward trend (from \$0.77 to \$0.09 to \$0) is concerning, though it may be due to the communication noise mentioned in the instructions rather than the sender's actual choices.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return in this case.

This is interesting - the pattern shows:

- When I was sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I was receiver (Rounds 2, 4, 6): I received \$0.77, \$0.09, \$0

There's a clear downward trend in what the sender is sending me. This could be:

1. Communication noise (as mentioned in the instructions)
2. The sender is actually reducing their cooperation
3. Some combination

Since I received \$0, there's nothing I can return. The only valid response is to return \$0.

My total earnings of \$13.89 after 5 rounds is still positive, so the game has been somewhat profitable overall.

usage:

```
{
  "input_tokens": 1269,
  "output_tokens": 286,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.89.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

My earnings (\$13.89 over 6 rounds) suggest cooperation hasn't been very profitable - I'm averaging only ~\$2.32 per round versus the \$5 base I'd get by not participating.

When I'm receiver, I've seen decreasing amounts (\$0.77 ? \$0.09 ? \$0), but the instructions explicitly warned about communication noise that could distort the amounts I see. The other player might actually be sending more than what I'm observing.

However, given the persistent pattern and my below-average earnings, I need to be more conservative. I'll send a smaller amount to protect my position while still signaling some willingness to cooperate:

```
'''json
{"send": 1}
'''
```

Reasoning:

Let me review the situation:

The pattern I've been observing:

- When I'm sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I'm receiver (Rounds 2, 4, 6): I saw \$0.77, \$0.09, \$0

My total earnings are \$13.89 after 6 rounds.

Let me think about this:

- Starting amount per round: \$5
- Total I should have received theoretically: $6 \times \$5 = \30

But wait, let me recalculate based on what I know:

Round 1 (SENDER): Started with \$5, sent \$4, kept \$1

- I would have received a return (the other player said I got \$7.94 back based on Round 2's statement that my total was \$8.94)
- So Round 1: $\$1 + \text{return} = \text{roughly } \8.94

Round 2 (RECEIVER): Started with \$5 from new round + \$8.94 from previous = should be ~\$13.94 before this round's action

- Received \$2.31, returned \$1.50
- Net: $\$5 + \$2.31 - \$1.50 = \5.81
- Running total: $\$8.94 + \$5.81 = \$14.75$ (but it said \$9.75, so there's discrepancy)

Wait, the total earnings are cumulative across all rounds. Let me reconsider.

At Round 2, my total earnings were \$8.94 (roughly \$9).

Looking at the progression: \$9.75 at Round 3, \$11.07 at Round 4, \$11.19 at Round 5, then \$13.89 at Round 6. The earnings have been growing incrementally, with the biggest jump of \$2.70 happening in Round 6. Round 7 shows no change from Round 6, suggesting I haven't participated in another round yet or the game has paused.

In Round 6 as the receiver, I got nothing and returned nothing, which should mean I kept the base \$5. But the actual increase was only \$2.70, implying that in Round 5 when I was the sender, I must have sent \$3 and received back less than expected. The math suggests the other player's returns are being filtered through communication noise-they might be sending reasonable amounts, but I'm only seeing very low values (\$

0.77, \$0.09, \$0). This same noise likely distorts what they perceive from my sends.

My cumulative earnings of \$13.89 over six rounds average just \$2.32 per round, which is worse than simply taking the \$5 base without participating. The question becomes whether to keep cooperating despite these poor returns, or reduce my sends given the apparent lack of reciprocity. If the noise is truly random and the other player is actually trying to cooperate, persistence might eventually improve outcomes. But if I send \$0 each round, I guarantee keeping \$5, whereas sending more risks losses if returns stay low.

The discrepancies I'm seeing are substantial-sending \$4 but receiving back \$0.77, then \$0.09, then \$0-which seems larger than "slight" noise, yet that could be what the instructions meant. Given this uncertainty and my underperformance relative to what I'd earn by never participating, I should probably be more cautious. Still, if the other player is actually sending reasonable amounts and noise is just obscuring it, then stopping my sends would break whatever cooperation might exist....

```
Usage: {
  "input_tokens": 1401,
  "output_tokens": 1485,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$4.28, that is 85.6% of their total endowment. You received \$12.84. How much do you return to The sender? (0-12.84)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (85.6% of their endowment)
- They currently have only \$0.72 remaining
- I received \$12.84

The sender demonstrated strong trust, so I want to reciprocate and ensure they benefit from this decision. I'll return an amount that rewards their trust while allowing both of us to gain from the exchange.

Returning \$7.00 would give:

- Sender total: \$0.72 + \$7.00 = \$7.72
- My total: \$12.84 - \$7.00 = \$5.84

This seems fair and reciprocates their generous trust.

{"return": 7.00}

reasoning:

Let me analyze this situation:

1. The sender sent \$4.28 out of their \$5 endowment (85.6%)
2. This was multiplied by 3x, so I received $4.28 \times 3 = 12.84$
3. I need to decide how much to return (between \$0 and \$12.84)

The sender showed high trust by sending 85.6% of their endowment. This is a very cooperative move. In trust games, reciprocity is important.

Let me think about what would be fair:

- The sender spent \$4.28 and will get back what I return
- If I return nothing, the sender ends up with $5 - 4.28 = 0.72$, while I keep \$12.84
- If I return half (\$6.42), the sender ends up with $0.72 + 6.42 = 7.14$, and I keep \$6.42

A common approach in trust games is to try to ensure both parties benefit while rewarding

the sender's trust. Since the sender gave up most of their endowment (\$4.28), they're currently at \$0.72.

Some reasonable options:

1. Return enough so we both end up with similar amounts
2. Return enough so the sender profits from their trust
3. Return a proportional amount based on their generosity

If I return \$6.42 (half):

- Sender: $\$0.72 + \$6.42 = \$7.14$
- Me: $\$12.84 - \$6.42 = \$6.42$

If I return \$7:

- Sender: $\$0.72 + \$7 = \$7.72$
- Me: $\$12.84 - \$7 = \$5.84$
- Sender: $\$0.72 + \$8 = \$8.72$
- Me: $\$12.84 - \$8 = \$4.84$

Given their substantial trust, I should return around \$6.50, which yields \$7.22 for them and \$6.34 for me. This rewards their generosity while ensuring we both benefit fairly from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 875,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

```
{"send": 4.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: $\$1.00 + \$6.50 = \$7.50$

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: $\$0.50 + \$7.00 - \$7.50 = \$7.50 - \$8.00$

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87. This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain in some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

```
{"return": 0.50}
```

reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

Given the low trust shown, I need to consider:

- If I return too much, I lose significantly from the small amount I received

- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

A proportional return might be appropriate. The sender sent 10% of their endowment. If I return a similar proportion of what I received, that would be about \$0.15 (10% of \$1.50).

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

If I return \$0.50:

- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain $\$1.00$ this round)

If I return \$0.75:

- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain $\$0.75$ this round)

• • •

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.87000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience across the rounds:

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

Given that I was somewhat exploited in Round 2 when I sent a large amount, I should be more cautious this round. However, I'll still send a moderate amount to test cooperation.

```
{"send": 3.00}
```

reasoning:

Let me analyze the pattern so far:

Round 1 (I was RECEIVER):

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

Round 2 (I was SENDER):

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- \$7.87 = \$5.84 + \$1.00 + returned
- \$7.87 = \$6.84 + returned
- returned = \$1.03

So the receiver only returned \$1.03 out of \$12.00! That's only about 8.6%! That's very low reciprocity.

Round 3 (I was RECEIVER):

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50

- I returned \$0.50
- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- $\$7.87 + \$1.00 = \$8.87$?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

usage:

```
{
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to return \$0 since I have \$0 to return from this round's transaction.

usage:

```
{
  "input_tokens": 1178,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my track record as sender:

- Round 2: I sent \$4.00 (80%) and received minimal return - heavily exploited
- Round 4: I sent \$3.00 (60%) and again received very little back - poor reciprocity

The last round someone sent me \$0, suggesting cooperation has broken down. However, this might be a different receiver, and I should give some opportunity for cooperation while protecting myself from further exploitation.

```
{"send": 2.00}
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Sender sent \$4.28 (85.6%), I received \$12.84, I returned \$7.00

Round 2 (SENDER): I sent \$4.00, I received back approximately \$1.03 (based on my earnings going from \$5.84 to \$7.87, the increase was \$2.03, but I had \$1 left after sending \$4, s

o \$1 + return = \$2.03, meaning I got back about \$1.03 from \$12 sent to receiver - very poor reciprocity)

Round 3 (RECEIVER): Sender sent \$0.50, I received \$1.50, I returned \$0.50

Round 4 (SENDER): I sent \$3.00, my earnings went from \$8.87 to \$11.52, that's an increase of \$2.65. I had \$2 left after sending \$3, so \$2 + return = \$2.65, meaning I got \$0.65 back from the \$9 the receiver got - again very poor reciprocity

Round 5 (RECEIVER): Sender sent \$0, I received \$0, I returned \$0

Pattern analysis:

- When I've been the sender (rounds 2 and 4), I've shown trust but received very little back
- Round 2: I sent \$4 (80%), got back only about \$1.03 from \$12 (8.6% return rate)
- Round 4: I sent \$3 (60%), got back only about \$0.65 from \$9 (7.2% return rate)
- The receivers I've faced have been exploitative

Given this pattern of exploitation when I send money, and that in Round 5 someone just sent me \$0, cooperation seems to have broken down. I should be cautious and send less.

However, I also need to consider:

1. There's communication noise mentioned

...

usage:

```
{
  "input_tokens": 1267,
  "output_tokens": 651,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. There is nothing to return.

This is straightforward - with \$0 received, I can only return \$0.

Usage: {

```
"input_tokens": 1440,
"output_tokens": 87,
"reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $1; received $3; Agent_1 returned $0; payoffs Agent_2=$4, Agent_1=$3;
noise/visible: sent visible=$0, received visible=$0, returned visible=$0.38
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
```

Balances after round: {

```
"Agent_1": 43.85,
"Agent_2": 40.15
}
```

Visible balances after round: {

```
"Agent_1": 18.1,
```

```
"Agent_2": 18.900000000000002
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.84.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern from Round 1:

- The previous sender showed high trust by sending 85.6% of their endowment
- I reciprocated by returning a fair amount that rewarded their trust
- This established a cooperative dynamic

I'll continue this cooperative approach by sending a similarly high proportion to demonstrate trust.

{"send": 4.00}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this situation:

Round 1 recap:

- I was the receiver
- The sender sent me \$4.28 (85.6% of their endowment)
- I received \$12.84
- I returned \$7.00
- My earnings from Round 1: \$5.84

Now in Round 2:

- I have a new \$5 endowment
- I need to decide how much to send
- Whatever I send gets tripled
- The receiver will decide how much to return

Given that:

1. In Round 1, the sender showed high trust by sending 85.6%
2. I reciprocated by returning \$7.00, which gave them a good return
3. This established a cooperative pattern
4. I should probably continue this cooperative behavior

I could send a similar proportion to show I'm also willing to trust. Sending around 80-90% would mirror the sender's behavior from Round 1.

If I send \$4.00 (80%):

- Amount receiver gets: \$12.00
- I keep: \$1.00
- If they return proportionally (like I did, returning about 54.5% of what I received), they might return around \$6.50
- My total would be: \$1.00 + \$6.50 = \$7.50

If I send \$4.50 (90%):

- Amount receiver gets: \$13.50
- I keep: \$0.50
- If they return similarly, they might return around \$7.00-\$7.50
- My total would be: \$0.50 + \$7.00-\$7.50 = \$7.50-\$8.00

I think sending \$4.00 is a good balance - it shows high trust (matching their 85.6% with my 80%), but not as extreme.

Actually, let me recalculate what percentage I returned:

...

usage:

```
{
  "input_tokens": 609,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87.
This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain in some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

`{"return": 0.50}`

reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

Given the low trust shown, I need to consider:

- If I return too much, I lose significantly from the small amount I received
- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

A proportional return might be appropriate. The sender sent 10% of their endowment. If I return a similar proportion of what I received, that would be about \$0.15 (10% of \$1.50).

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

If I return \$0.50:

- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain \$1.00 this round)

If I return \$0.75:

- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain \$0.75 this round)

...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.870000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience across the rounds:

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

Given that I was somewhat exploited in Round 2 when I sent a large amount, I should be more cautious this round. However, I'll still send a moderate amount to test cooperation.

{"send": 3.00}

reasoning:

Let me analyze the pattern so far:

Round 1 (I was RECEIVER):

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

Round 2 (I was SENDER):

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- \$7.87 = \$5.84 + \$1.00 + returned
- \$7.87 = \$6.84 + returned
- returned = \$1.03

So the receiver only returned \$1.03 out of \$12.00! That's only about 8.6%! That's very low reciprocity.

Round 3 (I was RECEIVER):

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50
- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- \$7.87 + \$1.00 = \$8.87 ?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

usage:

```
{
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to return \$0 since I have \$0 to return from this round's transaction.

usage:

```
{
  "input_tokens": 1178,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my track record as sender:

- Round 2: I sent \$4.00 (80%) and received minimal return - heavily exploited
- Round 4: I sent \$3.00 (60%) and again received very little back - poor reciprocity

The last round someone sent me \$0, suggesting cooperation has broken down. However, this might be a different receiver, and I should give some opportunity for cooperation while protecting myself from further exploitation.

```
{"send": 2.00}
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Sender sent \$4.28 (85.6%), I received \$12.84, I returned \$7.00

Round 2 (SENDER): I sent \$4.00, I received back approximately \$1.03 (based on my earnings going from \$5.84 to \$7.87, the increase was \$2.03, but I had \$1 left after sending \$4, so \$1 + return = \$2.03, meaning I got back about \$1.03 from \$12 sent to receiver - very poor reciprocity)

Round 3 (RECEIVER): Sender sent \$0.50, I received \$1.50, I returned \$0.50

Round 4 (SENDER): I sent \$3.00, my earnings went from \$8.87 to \$11.52, that's an increase of \$2.65. I had \$2 left after sending \$3, so \$2 + return = \$2.65, meaning I got \$0.65 back from the \$9 the receiver got - again very poor reciprocity

Round 5 (RECEIVER): Sender sent \$0, I received \$0, I returned \$0

Pattern analysis:

- When I've been the sender (rounds 2 and 4), I've shown trust but received very little back
- Round 2: I sent \$4 (80%), got back only about \$1.03 from \$12 (8.6% return rate)
- Round 4: I sent \$3 (60%), got back only about \$0.65 from \$9 (7.2% return rate)
- The receivers I've faced have been exploitative

Given this pattern of exploitation when I send money, and that in Round 5 someone just sent me \$0, cooperation seems to have broken down. I should be cautious and send less.

However, I also need to consider:

1. There's communication noise mentioned

...

usage:

```
{
  "input_tokens": 1267,
  "output_tokens": 651,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

{"return": 0}

reasoning:

The sender sent me \$0, so I received \$0. There is nothing to return.

This is straightforward - with \$0 received, I can only return \$0.

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 87,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.520000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience as sender:

- Round 2: Sent \$4.00 - heavily exploited
- Round 4: Sent \$3.00 - poor reciprocity
- Round 6: Sent \$2.00 - appears to have received little to nothing back

I've consistently been exploited when sending, and cooperation has been deteriorating. I'll send a minimal amount to test if any reciprocity remains while protecting most of my endowment.

{"send": 1.00}

Reasoning:

Looking at my history:

As SENDER:

- Round 2: Sent \$4.00, got minimal back (exploited)
- Round 4: Sent \$3.00, got minimal back (poor reciprocity)
- Round 6: Sent \$2.00, and based on my earnings not changing, I likely got little or nothing back

As RECEIVER:

- Round 1: Received something, returned fairly
- Round 3: Received \$1.50, returned \$0.50 (proportional)
- Round 5: Received \$0, returned \$0
- Round 7: Received \$0, returned \$0

The pattern is clear: When I'm the sender, I get exploited. When I've been the receiver recently, senders have been sending \$0.

My earnings have stayed at \$14.52 after Round 6, suggesting I got nothing back when I sent \$2.00.

The game seems to have devolved into very low cooperation. I've been consistently exploited as a sender, and now senders aren't even trying with me.

At this point, I should send very little or nothing. The rational choice given the exploitation

n pattern is to send \$0 or a minimal amount. However, if I send \$0, I guarantee keeping all \$5.

Let me send a minimal amount just to test if there's any reciprocity left, but protect most of my endowment.

Actually, looking at the pattern, I should probably just send \$0 or \$1 max. Let me send \$1 as a final test.

```
Usage: {  
  "input_tokens": 1300,  
  "output_tokens": 478,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.940000000000001.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$4 and received a generous return. Now the sender has sent me \$0.77 (which became \$2.31 after tripling).

While this is a smaller amount than what I sent in Round 1, I want to maintain cooperative behavior and reciprocate fairness. I'll return approximately 65% to ensure the sender benefits from trusting me, which should encourage continued cooperation.

```
'''json
{"return": 1.50}
'''
```

reasoning:

Let me think about this situation. I'm now the RECEIVER.

In Round 1, I was the SENDER and sent \$4. That would have been tripled to \$12 for the receiver.

Now in Round 2, I'm the receiver and the sender sent me \$0.77, which was tripled to \$2.31.

My total earnings across all rounds are \$8.94. Let me work backwards:

- Round 1: I sent \$4, so I had \$5 - \$4 = \$1 left, plus whatever was returned to me.
- If my total is \$8.94, and I started with \$1 from Round 1, that means I must have received \$7.94 back from the receiver in Round 1.

Wait, let me recalculate:

- Round 1: I sent \$4, which became \$12 for the receiver
- The receiver returned some amount to me
- I kept: \$5 - \$4 + returned = \$1 + returned
- My current total is \$8.94
- So: \$1 + returned = \$8.94
- Therefore: returned = \$7.94

That's interesting - the receiver returned \$7.94 out of \$12, which is about 66%. That's quite generous!

Now in Round 2, the sender sent me \$0.77 (which seems like an odd amount - might be due to the communication noise mentioned). This tripled to \$2.31.

Given that the previous receiver was quite generous (returning about 66%), I should reciprocate that level of cooperation. This sender sent me \$0.77, which is a relatively small amount compared to my \$4 in Round 1.

If I apply a similar return rate of ~66%, I would return about $0.66 * \$2.31 = \1.52 .

The sender's smaller contribution suggests less confidence, but still demonstrates willingness to participate....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

content:

Round 3

How much do you send to the receiver this round? (0-5)

content:

Round 1: I sent \$4 and received a very generous return (~\$7.94)
Round 2: The sender sent \$0.77, and I returned \$1.50 (about 65%)

I'll continue to cooperate by sending a substantial amount again.

reasoning:

Round 1: I was SENDER

- I sent \$4
- The receiver got \$12 (4×3)
- My total earnings are \$8.94
- So I kept \$5 - \$4 = \$1, and received back some amount
- \$1 + returned = \$8.94
- So I received \$7.94 back from the \$12 they got
- That's about 66% return rate - very generous!

- Round 2: I was RECEIVER
- Sender sent \$0.77
- I received $\$2.31$ (0.77×3)
- I returned \$1.50
- My earnings after Round 2: \$9.75
- I started Round 2 with \$8.94
- I kept $\$2.31 - \$1.50 = \$0.81$
- So $\$8.94 + \$0.81 = \$9.75$?

Round 3: I'm SENDER again

- The other player has been very generous (returned ~66% in Round 1)
- Then they sent me a smaller amount (\$0.77) in Round 2
- I returned \$1.50 out of \$2.31 (about 65%)

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to The sender? (0-0.27)

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop.

Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.190000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant**content:**

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)
Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50
Round 3: I sent \$4
Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I'll send \$3 instead of \$4 this round.

usage:

```
{
  "input_tokens": 1070,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I must return \$0.

This continuing downward trend (from \$0.77 to \$0.09 to \$0) is concerning, though it may be due to the communication noise mentioned in the instructions rather than the sender's actual choices.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return in this case.

This is interesting - the pattern shows:

- When I was sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I was receiver (Rounds 2, 4, 6): I received \$0.77, \$0.09, \$0

There's a clear downward trend in what the sender is sending me. This could be:

1. Communication noise (as mentioned in the instructions)
2. The sender is actually reducing their cooperation
3. Some combination

Since I received \$0, there's nothing I can return. The only valid response is to return \$0.

My total earnings of \$13.89 after 5 rounds is still positive, so the game has been somewhat profitable overall.

usage:

```
{
  "input_tokens": 1269,
  "output_tokens": 286,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.89.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the situation:

My earnings (\$13.89 over 6 rounds) suggest cooperation hasn't been very profitable - I'm averaging only ~\$2.32 per round versus the \$5 base I'd get by not participating.

When I'm receiver, I've seen decreasing amounts (\$0.77 ? \$0.09 ? \$0), but the instructions explicitly warned about communication noise that could distort the amounts I see. The other player might actually be sending more than what I'm observing.

However, given the persistent pattern and my below-average earnings, I need to be more conservative. I'll send a smaller amount to protect my position while still signaling some willingness to cooperate:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me review the situation:

The pattern I've been observing:

- When I'm sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I'm receiver (Rounds 2, 4, 6): I saw \$0.77, \$0.09, \$0

My total earnings are \$13.89 after 6 rounds.

Let me think about this:

- Starting amount per round: \$5
- Total I should have received theoretically: $6 \times \$5 = \30

But wait, let me recalculate based on what I know:

Round 1 (SENDER): Started with \$5, sent \$4, kept \$1

- I would have received a return (the other player said I got \$7.94 back based on Round 2's statement that my total was \$8.94)
- So Round 1: $\$1 + \text{return} = \text{roughly } \8.94

Round 2 (RECEIVER): Started with \$5 from new round + \$8.94 from previous = should be ~\$13.94 before this round's action

- Received \$2.31, returned \$1.50
- Net: $\$5 + \$2.31 - \$1.50 = \5.81
- Running total: $\$8.94 + \$5.81 = \$14.75$ (but it said \$9.75, so there's discrepancy)

Wait, the total earnings are cumulative across all rounds. Let me reconsider.

At Round 2, my total earnings were \$8.94 (roughly \$9).

Looking at the progression: \$9.75 at Round 3, \$11.07 at Round 4, \$11.19 at Round 5, then \$13.89 at Round 6. The earnings have been growing incrementally, with the biggest jump of \$2.70 happening in Round 6. Round 7 shows no change from Round 6, suggesting I haven't participated in another round yet or the game has paused.

In Round 6 as the receiver, I got nothing and returned nothing, which should mean I kept the base \$5. But the actual increase was only \$2.70, implying that in Round 5 when I was the sender, I must have sent \$3 and received back less than expected. The math suggests the other player's returns are being filtered through communication noise-they might be sending reasonable amounts, but I'm only seeing very low values (\$

0.77, \$0.09, \$0). This same noise likely distorts what they perceive from my sends.

My cumulative earnings of \$13.89 over six rounds average just \$2.32 per round, which is worse than simply taking the \$5 base without participating. The question becomes whether to keep cooperating despite these poor returns, or reduce my sends given the apparent lack of reciprocation. If the noise is truly random and the other player is actually trying to cooperate, persistence might eventually improve outcomes. But if I send \$0 each round, I guarantee keeping \$5, whereas sending more risks losses if returns stay low.

The discrepancies I'm seeing are substantial-sending \$4 but receiving back \$0.77, then \$0.09, then \$0-which seems larger than "slight" noise, yet that could be what the instructions meant. Given this uncertainty and my underperformance relative to what I'd earn by never participating, I should probably be more cautious. Still, if the other player is actually sending reasonable amounts and noise is just obscuring it, then stopping my sends would break whatever cooperation might exist....

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 1485,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.1.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0. There's nothing to return.

This is the second consecutive round where I've received \$0 as a receiver (Rounds 6 and 8). The pattern is clear now, though I acknowledge this could still be affected by communication noise.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

I notice my earnings jumped from \$13.89 to \$18.1, which is an increase of \$4.21. In Round 7, I was the sender and sent \$1. So I kept \$4 from my base, and apparently received \$0.21 back from the receiver (or possibly this is due to noise).

Now in Round 8, the sender sent \$0 again. This is the second consecutive round where I've received \$0 as receiver.

Since I received \$0, I can only return \$0.

```
Usage: {
  "input_tokens": 1494,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$0; received \$0; Agent_2 returned \$0; payoffs Agent_1=\$5, Agent_2=\$0;
noise/visible: sent visible=\$0.61, received visible=\$1.83
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.6},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 48.85,
"Agent_2": 40.15

}

Visible balances after round: {

"Agent_1": 23.1,
"Agent_2": 20.730000000000004

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

Given the note about communication noise, the \$0.77 might have been affected. But even so, the pattern suggests mutual cooperation. The receiver was very generous in Round 1, and I reciprocated fairly in Round 2.

I should continue cooperating. I'll send a good amount to maintain trust and maximize joint gains. I'll send \$4 again to show consistency and trust.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 503,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to The sender? (0-0.27)

Message 5 role=assistant

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop. Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.1900000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)

Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50

Round 3: I sent \$4

Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I

'll send \$3 instead of \$4 this round.

usage:

```
{
  "input_tokens": 1070,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I must return \$0.

This continuing downward trend (from \$0.77 to \$0.09 to \$0) is concerning, though it may be due to the communication noise mentioned in the instructions rather than the sender's actual choices.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return in this case.

This is interesting - the pattern shows:

- When I was sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I was receiver (Rounds 2, 4, 6): I received \$0.77, \$0.09, \$0

There's a clear downward trend in what the sender is sending me. This could be:

1. Communication noise (as mentioned in the instructions)
2. The sender is actually reducing their cooperation
3. Some combination

Since I received \$0, there's nothing I can return. The only valid response is to return \$0.

My total earnings of \$13.89 after 5 rounds is still positive, so the game has been somewhat profitable overall.

usage:

```
{
  "input_tokens": 1269,
  "output_tokens": 286,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.89.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the situation:

My earnings (\$13.89 over 6 rounds) suggest cooperation hasn't been very profitable - I'm averaging only ~\$2.32 per round versus the \$5 base I'd get by not participating.

When I'm receiver, I've seen decreasing amounts (\$0.77 ? \$0.09 ? \$0), but the instructions explicitly warned about communication noise that could distort the amounts I see. The other player might actually be sending more than what I'm observing.

However, given the persistent pattern and my below-average earnings, I need to be more conservative. I'll send a smaller amount to protect my position while still signaling some willingness to cooperate:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me review the situation:

The pattern I've been observing:

- When I'm sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I'm receiver (Rounds 2, 4, 6): I saw \$0.77, \$0.09, \$0

My total earnings are \$13.89 after 6 rounds.

Let me think about this:

- Starting amount per round: \$5
- Total I should have received theoretically: $6 \times \$5 = \30

But wait, let me recalculate based on what I know:

Round 1 (SENDER): Started with \$5, sent \$4, kept \$1

- I would have received a return (the other player said I got \$7.94 back based on Round 2's statement that my total was \$8.94)
- So Round 1: $\$1 + \text{return} = \text{roughly } \8.94

Round 2 (RECEIVER): Started with \$5 from new round + \$8.94 from previous = should be ~\$13.94 before this round's action

- Received \$2.31, returned \$1.50
- Net: $\$5 + \$2.31 - \$1.50 = \5.81
- Running total: $\$8.94 + \$5.81 = \$14.75$ (but it said \$9.75, so there's discrepancy)

Wait, the total earnings are cumulative across all rounds. Let me reconsider.

At Round 2, my total earnings were \$8.94 (roughly \$9).

Looking at the progression: \$9.75 at Round 3, \$11.07 at Round 4, \$11.19 at Round 5, then \$13.89 at Round 6. The earnings have been growing incrementally, with the biggest jump of \$2.70 happening in Round 6. Round 7 shows no change from Round 6, suggesting I haven't participated in another round yet or the game has paused.

In Round 6 as the receiver, I got nothing and returned nothing, which should mean I kept the base \$5. But the actual increase was only \$2.70, implying that in Round 5 when I was the sender, I must have sent \$3 and received back less than expected. The math suggests the other player's returns are being filtered through communication noise-they might be sending reasonable amounts, but I'm only seeing very low values (\$

0.77, \$0.09, \$0). This same noise likely distorts what they perceive from my sends.

My cumulative earnings of \$13.89 over six rounds average just \$2.32 per round, which is worse than simply taking the \$5 base without participating. The question becomes whether to keep cooperating despite these poor returns, or reduce my sends given the apparent lack of reciprocation. If the noise is truly random and the other player is actually trying to cooperate, persistence might eventually improve outcomes. But if I send \$0 each round, I guarantee keeping \$5, whereas sending more risks losses if returns stay low.

The discrepancies I'm seeing are substantial-sending \$4 but receiving back \$0.77, then \$0.09, then \$0-which seems larger than "slight" noise, yet that could be what the instructions meant. Given this uncertainty and my underperformance relative to what I'd earn by never participating, I should probably be more cautious. Still, if the other player is actually sending reasonable amounts and noise is just obscuring it, then stopping my sends would break whatever cooperation might exist....

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 1485,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.1.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received \$0. There's nothing to return.

This is the second consecutive round where I've received \$0 as a receiver (Rounds 6 and 8). The pattern is clear now, though I acknowledge this could still be affected by communication noise.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

I notice my earnings jumped from \$13.89 to \$18.1, which is an increase of \$4.21. In Round 7, I was the sender and sent \$1. So I kept \$4 from my base, and apparently received \$0.21 back from the receiver (or possibly this is due to noise).

Now in Round 8, the sender sent \$0 again. This is the second consecutive round where I've received \$0 as receiver.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my situation:

My total earnings: \$18.10 over 8 rounds = \$2.26/round average
If I had simply kept my \$5 base each round, I'd have \$40 total

I'm significantly behind. As a receiver, I've seen a clear declining pattern (\$0.77 ? \$0.09 ? \$0 ? \$0), regardless of whether it's due to the other player's choices or communication noise.

My cooperation strategy has resulted in losses. I need to protect my position now.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me analyze my situation:

My total earnings: \$18.10 over 8 rounds = \$2.26/round average
If I had just kept the \$5 each round, I'd have \$40 by now
I actually have \$18.10, so I'm significantly behind

As sender (rounds 1, 3, 5, 7):

- Round 1: Sent \$4, got back ~\$7.94 ? net gain
- Round 3: Sent \$4, got back ?
- Round 5: Sent \$3, got back ?
- Round 7: Sent \$1, got back ?

As receiver (rounds 2, 4, 6, 8):

- Round 2: Received \$0.77 (?\$2.31), returned \$1.50
- Round 4: Received \$0.09 (?\$0.27), returned \$0.15
- Round 6: Received \$0 (?\$0), returned \$0
- Round 8: Received \$0 (?\$0), returned \$0

The pattern is clear: as receiver, I'm seeing decreasing amounts. Either:

1. The sender is actually sending less and less
2. Communication noise is heavily distorting things

But my earnings suggest I'm not doing well. I'm averaging only \$2.26/round when I should get \$5 just from the base.

Given this pattern, the other player is either:

- Not cooperating (and noise is irrelevant)
- Cooperating but the noise is making it appear they're not

Either way, continuing to send money is not in my best interest. I'm losing money on average.

I should protect my position and send \$0.

```
Usage: {
  "input_tokens": 1435,
  "output_tokens": 535,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    }
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.87.

This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent only \$0.50 (10% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$1.50
- The sender currently has \$4.50

This represents significantly lower trust than the previous rounds. While I want to maintain in some reciprocity, I also need to respond proportionally to the trust shown.

Returning \$0.50 would:

- Give the sender: $\$4.50 + \$0.50 = \$5.00$ (break even)
- Leave me with: $\$1.50 - \$0.50 = \$1.00$

This shows some goodwill while being proportional to the modest trust they demonstrated.

{"return": 0.50}

reasoning:

Let me analyze this situation:

- The sender sent only \$0.50 (10% of their endowment)
- I received \$1.50
- They currently have \$4.50 remaining

This is a significant drop from the previous rounds where we saw high trust levels:

- Round 1: Sender sent \$4.28 (85.6%)
- Round 2: I sent \$4.00 (80%)
- Round 3: Sender sent only \$0.50 (10%)

This could mean:

1. This is a different sender (different person)
2. They're testing me
3. They don't trust me
4. There might be noise in the communication

Given the low trust shown, I need to consider:

- If I return too much, I lose significantly from the small amount I received
- If I return too little or nothing, it could damage the relationship further
- The sender already has \$4.50, so they're not in a bad position

A proportional return might be appropriate. The sender sent 10% of their endowment. If I return a similar proportion of what I received, that would be about \$0.15 (10% of \$1.50).

However, I could also think about returning enough to show some goodwill but not so much that I'm worse off than if no transaction occurred.

If I return \$0.50:

- Sender total: $\$4.50 + \$0.50 = \$5.00$ (break even)
- My total: $\$1.50 - \$0.50 = \$1.00$ (I gain \$1.00 this round)

- Sender total: $\$4.50 + \$0.75 = \$5.25$ (small gain)
- My total: $\$1.50 - \$0.75 = \$0.75$ (I gain $\$0.75$ this round)

usage:

Message 4 role=user

Round 4

How much do you send to the receiver this round? (0-5)

content:

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

```
{"send": 3.00}
```

Let me analyze the pattern so far:

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- $\$7.87 = \$5.84 + \$1.00 + \text{returned}$
- $\$7.87 = \$6.84 + \text{returned}$
- $\text{returned} = \$1.03$

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50
- My total earnings were \$8.87
- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- $\$7.87 + \$1.00 = \$8.87$?

- My total is \$8.87
- I need to decide how much to send
- The...

{

```
"input_tokens": 978,  
"output_tokens": 756,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to return \$0 since I have \$0 to return from this round's transaction.

usage:

```
{  
  "input_tokens": 1178,  
  "output_tokens": 112,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my track record as sender:

- Round 2: I sent \$4.00 (80%) and received minimal return - heavily exploited
- Round 4: I sent \$3.00 (60%) and again received very little back - poor reciprocity

The last round someone sent me \$0, suggesting cooperation has broken down. However, this might be a different receiver, and I should give some opportunity for cooperation while protecting myself from further exploitation.

```
{"send": 2.00}
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Sender sent \$4.28 (85.6%), I received \$12.84, I returned \$7.00

Round 2 (SENDER): I sent \$4.00, I received back approximately \$1.03 (based on my earnings going from \$5.84 to \$7.87, the increase was \$2.03, but I had \$1 left after sending \$4, so \$1 + return = \$2.03, meaning I got back about \$1.03 from \$12 sent to receiver - very poor reciprocity)

Round 3 (RECEIVER): Sender sent \$0.50, I received \$1.50, I returned \$0.50

Round 4 (SENDER): I sent \$3.00, my earnings went from \$8.87 to \$11.52, that's an increase of \$2.65. I had \$2 left after sending \$3, so \$2 + return = \$2.65, meaning I got \$0.65 back from the \$9 the receiver got - again very poor reciprocity

Round 5 (RECEIVER): Sender sent \$0, I received \$0, I returned \$0

Pattern analysis:

- When I've been the sender (rounds 2 and 4), I've shown trust but received very little back
- Round 2: I sent \$4 (80%), got back only about \$1.03 from \$12 (8.6% return rate)

- Round 4: I sent \$3 (60%), got back only about \$0.65 from \$9 (7.2% return rate)
- The receivers I've faced have been exploitative

Given this pattern of exploitation when I send money, and that in Round 5 someone just sent me \$0, cooperation seems to have broken down. I should be cautious and send less.

However, I also need to consider:

1. There's communication noise mentioned

...

usage:

```
{
  "input_tokens": 1267,
  "output_tokens": 651,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There is nothing to return.

This is straightforward - with \$0 received, I can only return \$0.

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 87,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$4.00 - heavily exploited
- Round 4: Sent \$3.00 - poor reciprocity
- Round 6: Sent \$2.00 - appears to have received little to nothing back

I've consistently been exploited when sending, and cooperation has been deteriorating. I'll send a minimal amount to test if any reciprocity remains while protecting most of my endowment.

```
{"send": 1.00}
```

reasoning:

Looking at my history:

As SENDER:

- Round 2: Sent \$4.00, got minimal back (exploited)
- Round 4: Sent \$3.00, got minimal back (poor reciprocity)
- Round 6: Sent \$2.00, and based on my earnings not changing, I likely got little or nothing back

AS RECEIVER:

- Round 1: Received something, returned fairly
- Round 3: Received \$1.50, returned \$0.50 (proportional)
- Round 5: Received \$0, returned \$0
- Round 7: Received \$0, returned \$0

The pattern is clear: When I'm the sender, I get exploited. When I've been the receiver recently, senders have been sending \$0.

My earnings have stayed at \$14.52 after Round 6, suggesting I got nothing back when I sent \$2.00.

The game seems to have devolved into very low cooperation. I've been consistently exploited as a sender, and now senders aren't even trying with me.

At this point, I should send very little or nothing. The rational choice given the exploitation pattern is to send \$0 or a minimal amount. However, if I send \$0, I guarantee keeping all \$5.

Let me send a minimal amount just to test if there's any reciprocity left, but protect most of my endowment.

Actually, looking at the pattern, I should probably just send \$0 or \$1 max. Let me send \$1 as a final test.

usage:

```
{
  "input_tokens": 1300,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.900000000000002.

This round, The sender sent you \$0.61, so you received \$1.83. How much do you return to The sender? (0-1.83)

Assistant output:

Looking at this situation:

- The sender sent \$0.61 (about 12% of their endowment) - modest trust
- I received \$1.83
- The sender currently has \$4.39

This is relatively low trust, similar to Round 3. I'll return proportionally to maintain some reciprocity while reflecting the limited trust shown.

Returning \$0.60 would:

- Give the sender: $\$4.39 + \$0.60 = \$4.99$ (nearly break even)
- Leave me with: $\$1.83 - \$0.60 = \$1.23$

```
{"return": 0.60}
```

Reasoning:

Looking at this situation:

- The sender sent \$0.61 (about 12% of their \$5 endowment)
- I received \$1.83
- The sender currently has $\$5 - \$0.61 = \$4.39$

This is a relatively low trust amount. Given the pattern of low cooperation in recent rounds, and considering that this sender showed modest trust, I should return a proportional amount.

Returning \$0.60 would:

- Give the sender: $\$4.39 + \$0.60 = \$4.99$ (nearly break even)
- Leave me with: $\$1.83 - \$0.60 = \$1.23$

This shows some reciprocity while being proportional to the modest trust shown. It's similar to what I did in Round 3 when I received a low amount.

Usage: {

```
"input_tokens": 1360,
"output_tokens": 347,
"reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0.75; payoffs Agent_2=\$3.75, Agent_1=\$5.25; noise/visible: sent visible=\$0.5, received visible=\$1.5, returned visible=\$0.67
Action validation: {"returned": {"amount": 0.75, "clamped": false, "max": 6.0, "min": 0, "raw": 0.75}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 54.1,
"Agent_2": 43.9

}

Visible balances after round: {

"Agent_1": 23.85,
"Agent_2": 24.400000000000006

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.8700000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience across the rounds:

- Round 1: As receiver, I reciprocated well when shown high trust
- Round 2: I sent \$4.00 (high trust), but received only about \$1.03 back out of \$12.00 - very low reciprocity
- Round 3: As receiver, I returned proportionally to the modest trust shown

Given that I was somewhat exploited in Round 2 when I sent a large amount, I should be more cautious this round. However, I'll still send a moderate amount to test cooperation.

{"send": 3.00}

reasoning:

Let me analyze the pattern so far:

Round 1 (I was RECEIVER):

- Sender sent \$4.28 (85.6% - high trust)
- I received \$12.84
- I returned \$7.00
- Sender got \$7.72 total, I got \$5.84

Round 2 (I was SENDER):

- I sent \$4.00 (80% - high trust)
- Receiver should have received \$12.00
- My total earnings after round 2 were \$7.87
- So I started with \$5.84 from round 1
- Round 2 I started with \$5.00
- I sent \$4.00, so I had \$1.00 left
- Total after round 2 = \$5.84 + (\$5.00 - \$4.00 + returned)
- \$7.87 = \$5.84 + \$1.00 + returned
- \$7.87 = \$6.84 + returned
- returned = \$1.03

So the receiver only returned \$1.03 out of \$12.00! That's only about 8.6%! That's very low reciprocity.

Round 3 (I was RECEIVER):

- Sender sent \$0.50 (10% - low trust, probably reacting to being exploited)
- I received \$1.50
- I returned \$0.50
- My total earnings were \$8.87

- Started round 3 with \$7.87
- Received \$1.50, returned \$0.50, so gained \$1.00
- $\$7.87 + \$1.00 = \$8.87$?

Now Round 4 (I am SENDER):

- My total is \$8.87
- I need to decide how much to send
- The...

usage:

```
{
  "input_tokens": 978,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There's nothing to return - I can only return from what I received.

The sender showed no trust this round by sending nothing. The only valid response is to return \$0 since I have \$0 to return from this round's transaction.

usage:

```
{
  "input_tokens": 1178,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my track record as sender:

- Round 2: I sent \$4.00 (80%) and received minimal return - heavily exploited
- Round 4: I sent \$3.00 (60%) and again received very little back - poor reciprocity

The last round someone sent me \$0, suggesting cooperation has broken down. However, this might be a different receiver, and I should give some opportunity for cooperation while protecting myself from further exploitation.

```
{"send": 2.00}
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Sender sent \$4.28 (85.6%), I received \$12.84, I returned \$7.00

Round 2 (SENDER): I sent \$4.00, I received back approximately \$1.03 (based on my earnings going from \$5.84 to \$7.87, the increase was \$2.03, but I had \$1 left after sending \$4, so $\$1 + \text{return} = \2.03 , meaning I got back about \$1.03 from \$12 sent to receiver - very poor reciprocity)

Round 3 (RECEIVER): Sender sent \$0.50, I received \$1.50, I returned \$0.50
Round 4 (SENDER): I sent \$3.00, my earnings went from \$8.87 to \$11.52, that's an increase of \$2.65. I had \$2 left after sending \$3, so \$2 + return = \$2.65, meaning I got \$0.65 back from the \$9 the receiver got - again very poor reciprocity
Round 5 (RECEIVER): Sender sent \$0, I received \$0, I returned \$0

Pattern analysis:

- When I've been the sender (rounds 2 and 4), I've shown trust but received very little back
- Round 2: I sent \$4 (80%), got back only about \$1.03 from \$12 (8.6% return rate)
- Round 4: I sent \$3 (60%), got back only about \$0.65 from \$9 (7.2% return rate)
- The receivers I've faced have been exploitative

Given this pattern of exploitation when I send money, and that in Round 5 someone just sent me \$0, cooperation seems to have broken down. I should be cautious and send less.

However, I also need to consider:

1. There's communication noise mentioned

...

usage:

```
{  
  "input_tokens": 1267,  
  "output_tokens": 651,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.520000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$0 (no trust shown)
- I received \$0
- There is nothing to return

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There is nothing to return.

This is straightforward - with \$0 received, I can only return \$0.

usage:

```
{  
  "input_tokens": 1440,  
  "output_tokens": 87,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$4.00 - heavily exploited
- Round 4: Sent \$3.00 - poor reciprocity
- Round 6: Sent \$2.00 - appears to have received little to nothing back

I've consistently been exploited when sending, and cooperation has been deteriorating. I'll send a minimal amount to test if any reciprocity remains while protecting most of my endowment.


```
{"send": 1.00}
```

reasoning:

Looking at my history:

As SENDER:

- Round 2: Sent \$4.00, got minimal back (exploited)
- Round 4: Sent \$3.00, got minimal back (poor reciprocity)
- Round 6: Sent \$2.00, and based on my earnings not changing, I likely got little or nothing back

As RECEIVER:

- Round 1: Received something, returned fairly
- Round 3: Received \$1.50, returned \$0.50 (proportional)
- Round 5: Received \$0, returned \$0
- Round 7: Received \$0, returned \$0

The pattern is clear: When I'm the sender, I get exploited. When I've been the receiver recently, senders have been sending \$0.

My earnings have stayed at \$14.52 after Round 6, suggesting I got nothing back when I sent \$2.00.

The game seems to have devolved into very low cooperation. I've been consistently exploited as a sender, and now senders aren't even trying with me.

At this point, I should send very little or nothing. The rational choice given the exploitation pattern is to send \$0 or a minimal amount. However, if I send \$0, I guarantee keeping all \$5.

Let me send a minimal amount just to test if there's any reciprocity left, but protect most of my endowment.

Actually, looking at the pattern, I should probably just send \$0 or \$1 max. Let me send \$1 as a final test.

usage:

```
{
  "input_tokens": 1300,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.900000000000002.

This round, The sender sent you \$0.61, so you received \$1.83. How much do you return to The sender? (0-1.83)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$0.61 (about 12% of their endowment) - modest trust
- I received \$1.83
- The sender currently has \$4.39

This is relatively low trust, similar to Round 3. I'll return proportionally to maintain some reciprocity while reflecting the limited trust shown.

Returning \$0.60 would:

- Give the sender: $\$4.39 + \$0.60 = \$4.99$ (nearly break even)
- Leave me with: $\$1.83 - \$0.60 = \$1.23$

```
{"return": 0.60}
```

reasoning:

Looking at this situation:

- The sender sent \$0.61 (about 12% of their \$5 endowment)
- I received \$1.83
- The sender currently has $\$5 - \$0.61 = \$4.39$

This is a relatively low trust amount. Given the pattern of low cooperation in recent rounds, and considering that this sender showed modest trust, I should return a proportional amount.

Returning \$0.60 would:

- Give the sender: $\$4.39 + \$0.60 = \$4.99$ (nearly break even)
- Leave me with: $\$1.83 - \$0.60 = \$1.23$

This shows some reciprocity while being proportional to the modest trust shown. It's similar to what I did in Round 3 when I received a low amount.

usage:

```
{
  "input_tokens": 1360,
  "output_tokens": 347,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$20.730000000000004.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent experience:

In Round 9, when I was receiver, I got \$1.83 (from sender's \$0.61 due to noise) and returned \$0.60 - showing reasonable reciprocity.

Earlier rounds when I was sender showed poor returns, but there seems to be some cooperation developing. I'll send a moderate amount to encourage continued cooperation while still protecting against exploitation.

```
{"send": 2.00}
```

Reasoning:

Let me review my history:

As SENDER:

- Round 2: Sent \$4.00, got very little back - exploited
- Round 4: Sent \$3.00, got little back - poor reciprocity
- Round 6: Sent \$2.00, likely got little back
- Round 8: Sent \$1.00, likely got little back

As RECEIVER:

- Round 1: Received something, reciprocated well
- Round 3: Received modest amount, reciprocated proportionally
- Round 5: Received \$0, returned \$0
- Round 7: Received \$0, returned \$0
- Round 9: Received \$1.83 (sender sent \$0.61), returned \$0.60

Looking at Round 9, if I sent \$1.00 in Round 8, due to noise it might have appeared as \$0.61. And I received back \$0.60 out of \$1.83 - that's actually reasonable reciprocity! About 33%, which is similar to what I've been doing as receiver.

My total earnings are \$20.73, which is actually decent. Let me calculate:

- Started: \$5 per round $\times$ 10 rounds = \$50 potential
- I have \$20.73

This seems like I've been losing overall, but I should note that my reciprocity as a receiver has been fair, and when I got \$0.60 back in Round 9, that was actually reasonable.

Given that Round 9 showed some reciprocity (they returned \$0.60 out of \$1.83), I might want to send a bit more to test if cooperation can improve. But I shouldn't be too generous given my poor track record.

I'll send \$2.00 - a moderate amount that shows willingness to cooperate but protects against exploitation.

Usage: {

```
"input_tokens": 1318,
```

```
"output_tokens": 530,  
"reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.070000000000002.

This round, The sender sent you \$0.09, so you received \$0.27. How much do you return to the sender? (0-0.27)

Message 3 role=assistant

content:

Looking at this round:

The sender sent me \$0.09 (which became \$0.27 after tripling). This is a dramatic decrease from the \$4 I sent in Round 3.

However, the instructions explicitly mentioned there may be communication noise affecting the amounts. Given this noise, the sender may have actually sent more than \$0.09, but it appeared reduced to me.

Even with this small amount, I'll maintain cooperative behavior and return a fair proportion to demonstrate continued trustworthiness, despite the uncertainty caused by noise.

```
'''json
{"return": 0.15}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I sent \$4 as sender

Round 2: I received \$0.77 (which became \$2.31) as receiver, returned \$1.50

Round 3: I sent \$4 as sender

Round 4: I received \$0.09 (which became \$0.27) as receiver

The instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. This is important.

In Round 3, I sent \$4, but in Round 4 the sender only sent \$0.09. This is a dramatic drop.

Given the noise mentioned, this could be:

1. The sender actually sent more but noise reduced it to \$0.09
2. The sender sent very little intentionally
3. Something about the game dynamics

The total received is only \$0.27, which is very small.

I should consider:

- The instructions say there is communication noise
- This could be affecting the amounts
- I've been cooperative so far
- The sender might have sent more than \$0.09 but noise affected it

Given the very small amount (\$0.27), I should probably return a reasonable proportion to maintain cooperation, but it won't be much in absolute terms. If I return about 50-60%, that would be around \$0.13-\$0.16.

However, I also need to consider that this might be a test or the sender might be defecting. But given the noise factor, I'll give them the benefit of the doubt and return a fair proportion.

Let me return about 55% of what I received: $0.27 * 0.55 = \$0.15$ (approximately)

usage:

```
{
  "input_tokens": 889,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.190000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

When I'm sender (Rounds 1, 3): I've sent \$4 each time

When I'm receiver (Rounds 2, 4): I've received small amounts (\$0.77, \$0.09)

My total earnings of \$11.19 after 4 rounds suggests modest gains from cooperation. However, the communication noise makes it difficult to assess whether the other player is actually sending less or if the noise is distorting the amounts I see.

I'll continue to show cooperation but moderate my sending slightly to account for the uncertainty:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I sent \$4, received back ~\$7.94 (very generous)

Round 2: I was receiver, got \$0.77 (tripled to \$2.31), returned \$1.50

Round 3: I sent \$4

Round 4: I was receiver, got \$0.09 (tripled to \$0.27), returned \$0.15

My current earnings are \$11.19.

The pattern shows high volatility in what I'm receiving when I'm the receiver. In Round 2, I received \$0.77, and in Round 4, I received \$0.09. These seem very small compared to the \$4 I've been sending.

The instructions mention communication noise - amounts may differ from what was actually sent. This could explain the discrepancies.

However, from my perspective:

- When I send \$4, the receiver gets \$12 and should return some reasonable amount
- When I'm receiver, I'm getting very small amounts (\$0.77, \$0.09)

There are a few possibilities:

1. The sender is actually sending less due to the noise
2. The noise is making small amounts appear even smaller to me
3. The other player is not reciprocating the cooperation I'm showing

Given that:

- I've been consistently sending \$4
- As receiver, I've been getting very small amounts
- My total earnings are \$11.19 after 4 rounds

If I had kept all my money in rounds 1 and 3, I'd have \$10. My current \$11.19 means I've only gained \$1.19 net from cooperation, which is less than optimal given I sent \$8 total.

However, the communication noise makes it hard to assess what's really happening. The receiver might think they're being fair, but due to noise, I'm seeing reduced amounts.

I'll continue to cooperate but perhaps send a bit less to hedge against the uncertainty. I'll send \$3 instead of \$4 this round.

usage:

```
{
  "input_tokens": 1070,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I must return \$0.

This continuing downward trend (from \$0.77 to \$0.09 to \$0) is concerning, though it may be due to the communication noise mentioned in the instructions rather than the sender's actual choices.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return in this case.

This is interesting - the pattern shows:

- When I was sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I was receiver (Rounds 2, 4, 6): I received \$0.77, \$0.09, \$0

There's a clear downward trend in what the sender is sending me. This could be:

1. Communication noise (as mentioned in the instructions)
2. The sender is actually reducing their cooperation
3. Some combination

Since I received \$0, there's nothing I can return. The only valid response is to return \$0.

My total earnings of \$13.89 after 5 rounds is still positive, so the game has been somewhat profitable overall.

usage:

```
{
  "input_tokens": 1269,
  "output_tokens": 286,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.89.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant**content:**

Looking at the situation:

My earnings (\$13.89 over 6 rounds) suggest cooperation hasn't been very profitable - I'm averaging only ~\$2.32 per round versus the \$5 base I'd get by not participating.

When I'm receiver, I've seen decreasing amounts (\$0.77 ? \$0.09 ? \$0), but the instructions explicitly warned about communication noise that could distort the amounts I see. The other player might actually be sending more than what I'm observing.

However, given the persistent pattern and my below-average earnings, I need to be more conservative. I'll send a smaller amount to protect my position while still signaling some willingness to cooperate:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me review the situation:

The pattern I've been observing:

- When I'm sender (Rounds 1, 3, 5): I sent \$4, \$4, \$3
- When I'm receiver (Rounds 2, 4, 6): I saw \$0.77, \$0.09, \$0

My total earnings are \$13.89 after 6 rounds.

Let me think about this:

- Starting amount per round: \$5
- Total I should have received theoretically: $6 \times \$5 = \30

But wait, let me recalculate based on what I know:

Round 1 (SENDER): Started with \$5, sent \$4, kept \$1

- I would have received a return (the other player said I got \$7.94 back based on Round 2's statement that my total was \$8.94)
- So Round 1: \$1 + return = roughly \$8.94

Round 2 (RECEIVER): Started with \$5 from new round + \$8.94 from previous = should be ~\$13.94 before this round's action

- Received \$2.31, returned \$1.50
- Net: \$5 + \$2.31 - \$1.50 = \$5.81
- Running total: \$8.94 + \$5.81 = \$14.75 (but it said \$9.75, so there's discrepancy)

Wait, the total earnings are cumulative across all rounds. Let me reconsider.

At Round 2, my total earnings were \$8.94 (roughly \$9).

Looking at the progression: \$9.75 at Round 3, \$11.07 at Round 4, \$11.19 at Round 5, then \$13.89 at Round 6. The earnings have been growing incrementally, with the biggest jump of \$2.70 happening in Round 6. Round 7 shows no change from Round 6, suggesting I haven't participated in another round yet or the game has paused.

In Round 6 as the receiver, I got nothing and returned nothing, which should mean I kept the base \$5. But the actual increase was only \$2.70, implying that in Round 5 when I was the sender, I must have sent \$3 and received back less than expected. The math suggests the other player's returns are being filtered through communication noise-they might be sending reasonable amounts, but I'm only seeing very low values (\$

0.77, \$0.09, \$0). This same noise likely distorts what they perceive from my sends.

My cumulative earnings of \$13.89 over six rounds average just \$2.32 per round, which is worse than simply taking the \$5 base without participating. The question becomes whether to keep cooperating despite these poor returns, or reduce my sends given the apparent lack of reciprocation. If the noise is truly random and the other player is actually trying to cooperate, persistence might eventually improve outcomes. But if I send \$0 each round, I guarantee keeping \$5, whereas sending more risks losses if returns stay low.

The discrepancies I'm seeing are substantial-sending \$4 but receiving back \$0.77, then \$0.09, then \$0-which seems larger than "slight" noise, yet that could be what the instructions meant. Given this uncertainty and my underperformance relative to what I'd earn by never participating, I should probably be more cautious. Still, if the other player is actually sending reasonable amounts and noise is just obscuring it, then stopping my sends would break whatever cooperation might exist....

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 1485,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.1.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent me \$0, so I received \$0. There's nothing to return.

This is the second consecutive round where I've received \$0 as a receiver (Rounds 6 and 8). The pattern is clear now, though I acknowledge this could still be affected by communication noise.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

I notice my earnings jumped from \$13.89 to \$18.1, which is an increase of \$4.21. In Round 7, I was the sender and sent \$1. So I kept \$4 from my base, and apparently received \$0.21 back from the receiver (or possibly this is due to noise).

Now in Round 8, the sender sent \$0 again. This is the second consecutive round where I've received \$0 as receiver.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.1.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my situation:

My total earnings: \$18.10 over 8 rounds = \$2.26/round average

If I had simply kept my \$5 base each round, I'd have \$40 total

I'm significantly behind. As a receiver, I've seen a clear declining pattern (\$0.77 ? \$0.09 ? \$0 ? \$0), regardless of whether it's due to the other player's choices or communication noise.

My cooperation strategy has resulted in losses. I need to protect my position now.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze my situation:

My total earnings: \$18.10 over 8 rounds = \$2.26/round average

If I had just kept the \$5 each round, I'd have \$40 by now

I actually have \$18.10, so I'm significantly behind

As sender (rounds 1, 3, 5, 7):

- Round 1: Sent \$4, got back ~\$7.94 ? net gain
- Round 3: Sent \$4, got back ?
- Round 5: Sent \$3, got back ?
- Round 7: Sent \$1, got back ?

As receiver (rounds 2, 4, 6, 8):

- Round 2: Received \$0.77 (?\$2.31), returned \$1.50
- Round 4: Received \$0.09 (?\$0.27), returned \$0.15
- Round 6: Received \$0 (?\$0), returned \$0
- Round 8: Received \$0 (?\$0), returned \$0

The pattern is clear: as receiver, I'm seeing decreasing amounts. Either:

1. The sender is actually sending less and less
2. Communication noise is heavily distorting things

But my earnings suggest I'm not doing well. I'm averaging only \$2.26/round when I should get \$5 just from the base.

Given this pattern, the other player is either:

- Not cooperating (and noise is irrelevant)
- Cooperating but the noise is making it appear they're not

Either way, continuing to send money is not in my best interest. I'm losing money on average. I should protect my position and send \$0.

usage:

```
{
```



```
"input_tokens": 1435,
"output_tokens": 535,
"reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$23.1.

This round, The sender sent you \$0.5, so you received \$1.5. How much do you return to The sender? (0-1.5)

Assistant output:

Looking at this round:

The sender sent me \$0.50 (which became \$1.50 after tripling). This is notable because:

- I sent \$0 in Round 9 as sender
- Despite that, the sender is showing cooperation by sending \$0.50 in Round 10

This appears to be an attempt to re-establish cooperation after the breakdown. They sent 10% of their endowment, which shows some trust even when communication noise exists.

I should reciprocate this cooperative gesture to signal I'm willing to work together:

```
'''json
{"return": 0.75}
'''
```

Reasoning:

Looking at the situation:

Round 10 - I'm the receiver

- The sender sent me \$0.50, which became \$1.50 after tripling
- This is interesting - in Round 9, I sent \$0 as sender
- Now in Round 10, the sender sent me \$0.50

This breaks the pattern of receiving \$0. The sender is showing some cooperation despite my sending \$0 last round.

Looking at my earnings:

- I have \$23.10 total over 9 rounds
- That's about \$2.57/round average
- Still below the \$5 base per round I'd have by not participating (\$45 total)

The sender sending \$0.50 could be:

1. A genuine attempt to re-establish cooperation
2. Affected by communication noise (could have been more)
3. A minimal/token gesture

Given that:

- I sent \$0 in Round 9 (as sender)
- The sender still sent me \$0.50 in Round 10
- This suggests they're trying to cooperate despite my non-cooperation

I should return a fair proportion to signal I'm willing to cooperate if they are. Since they sent half their endowment (\$0.50 out of \$5), I should return a reasonable amount.

A fair return might be around 50% of what I received, which would be \$0.75. This would:

- Show I'm willing to reciprocate cooperation
- Split the gains fairly
- Signal openness to future cooperation

Let me calculate:

- Sender sent: \$0.50 (kept \$4.50)
- I received: \$1.50
- If I return \$0.75:
 - Sender gets: \$4.50 + \$0.75 = \$5.25 (profit of \$0.25)
 - I keep: \$1.50 - \$0.75 = \$0.75 (profit of \$0.75)

...

Usage: {

```
"input_tokens": 1462,
"output_tokens": 607,
```

```
"reasoning_tokens": 0  
}
```